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A Union of Filesystems

Discover more about the Overlay filesystem, which is excellent for packing a large set of applications into a limited space.

If you have ever used *Unetbootin* to create a bootable USB, you may have noticed the following option:

'Space used to preserve files across reboots (Ubuntu only)'

What is different about Ubuntu? Actually, it is not so limited and can be used for most distributions derived from Debian or Ubuntu.

On the other hand, if you use *Fedora LiveUSB Creator*, you would have noticed the option:

'Persistent storage'

Both Ubuntu and Fedora offer persistent storage but use different mechanisms. I have found persistent storage immensely useful, especially for booting from diskless machines. It is interesting to learn about the differences between them and how they work.

OverlayFS

If you try *Unetbootin*, it will create the file *casper-rw* on the USB. Besides, an additional kernel boot option—*persistent*—will be added in the Syslinux configuration file.

The persistent storage need not be a file. It can be a partition instead. The label of the partition has to be '*casper-rw*'. Persistence seems to work on a Mint live USB, provided a partition is used. For Mint,

- Create two partitions on a USB
- Install the live ISO on one partition
- Remove the *casper-rw* file
- Format the second partition as ext2
- Use *e2label* to change the label of the partition to '*casper-rw*'

The root system will be mounted as the '*overlays*' type, which is a module included in the Ubuntu/Debian derived distributions.

The system mounts the *squashfs* root as a read-only filesystem (*/rofs*). It then mounts the *casper-rw* file/partition as a read-write filesystem (*/cow*). Both are mounted as a composite filesystem with */rofs* as the lowerdir and */cow* as the upperdir. *Overlays* will search

for a file on upperdir first and then on lowerdir. If a file is modified, it will be stored in the upperdir.

Snapshots and device mapper

A layered filesystem is reminiscent of the description in Snapshot (computer storage) in Wikipedia. Volume Manager supports snapshots. Fedora does not include the *overlays* module and uses the concept of a snapshot to achieve the same goal.

As before, you will need a read-only filesystem. However, the overlay is no longer a filesystem. Instead, it stores information about the blocks that have been changed and the modified data of the block. So, the data read from the basic filesystem will be replaced by data in the modified blocks.

This has some interesting consequences. A Fedora live USB also contains a file with a squash filesystem. When you loop mount this file, you do not get a root filesystem. Instead, you will find that it contains an image of an ext4 filesystem '*LiveOS/ext3fs*'. The reason for this convoluted set-up becomes clear when you realise that *squashfs* is a read-only filesystem. Hence, you cannot use it as a read-only snapshot and modify its data.

Fedora LiveUSB Creator will create a file with the name *LiveOS/overlay-LIVE-<UUID>*. A kernel option — *rd.live.overlay=LABEL=LIVE* — would be added in the Syslinux configuration file. A script in the *initramfs* will loop mount the read-only ext4 filesystem. It will create a volume manager snapshot from it, and then mount the snapshot with the overlay file as a read-write filesystem on */dev/mapper/live-rw*. The essentials of the script will be as follows (adapted from *dmsquash-live-root.sh* in */usr/lib/dracut/modules.d/90dmsquash-live/*):

```
BASE=$( losetup -f )
losetup -r $BASE /run/initramfs/squashfs/LiveOS/ext3fs.
img
OVERLAY=$( losetup -f )
```